Image Generators are Generalist Vision Learners

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Research Questions

- Can image generation models serve as generalist vision learners by leveraging their learned visual representations for understanding tasks?
- How can vision tasks be reformulated as image generation problems to enable unified training and evaluation?
- Does lightweight instruction-tuning of generative models preserve generation capabilities while enabling SOTA vision understanding performance?

Core Idea

The paper hypothesizes that image generators, like language models, develop rich internal representations during pretraining that can be repurposed for understanding tasks. The key insight is to reformulate all vision tasks (segmentation, depth estimation, etc.) as RGB image generation problems - for example, generating segmentation masks as colored images where each color represents a class. This allows instruction-tuning a pre-trained image generator (Nano Banana Pro) with minimal vision task data to create Vision Banana, which can both generate images and perform vision understanding by producing decodable visualizations.

Key Formula

Depth mapping function

$$f(d, \lambda, c) = 1 - (1 - d_{\lambda c})^{\lambda + 1}$$

Maps normalized metric depth distances d to RGB color values for visualization. λ is the shape parameter (-3), c is the scale parameter (10/3), d_λc represents curved distances. This bijective mapping allows depth values to be encoded as RGB images and decoded back to metric depth for evaluation.

Key Figure

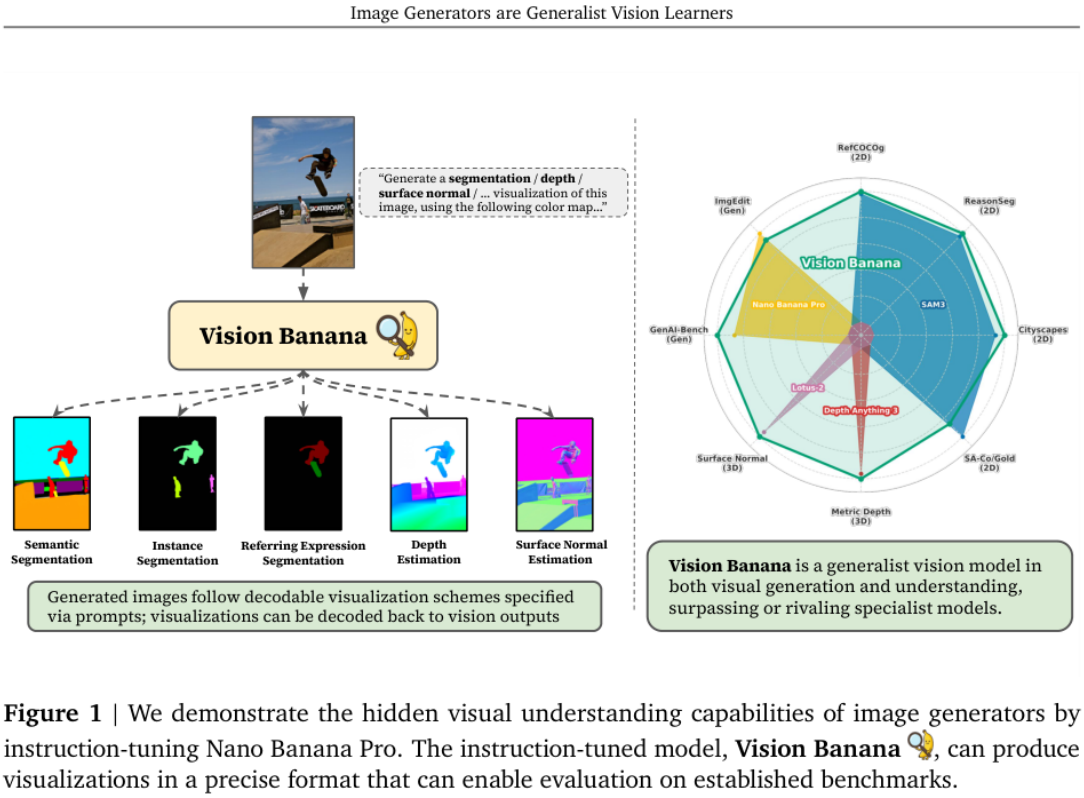

Figure 1: Shows the core concept of Vision Banana generating decodable visualizations for multiple vision tasks (segmentation, depth, surface normals) from natural language prompts, demonstrating the unified interface approach.

Limitations

- Requires careful design of bijective mappings between vision outputs and RGB representations for each task type
- Limited evaluation to specific vision tasks (segmentation, depth, surface normal estimation) - generalization to other domains unclear
- Computational overhead compared to lightweight specialist models - uses full generative model for understanding tasks
- Instance segmentation performance lags behind DINO-X, suggesting some tasks may not benefit equally from this approach